A1: Automation Landscape Report

Context and Rationale

Manufacturing Setting: The Ubuntu Naturals Johannesburg encapsulation flagship plant (South Africa, Tier 1; Stage 4 of the Ubuntu value chain; the group's MES anchor) is moving the Moringa Daily capsule line off its legacy standalone PLCs and onto an integrated MES/SCADA architecture. The site is SAHPRA-regulated and its finished product ships pan-African and into selected export markets, so whatever replaces those PLCs has to stand up to an inspection as well as to a shift. It also has to stand up to Eskom: load-shedding is routine here, and an architecture that cannot keep processing at the edge through an outage will not hold the line together.

Rationale: A SAHPRA inspector walking into an African pharmaceutical or wellness plant expects two things of its computerised systems: that they meet international data integrity standards, and that the automation underneath them supports validated, auditable production. AMRH-harmonised expectations mean the regional export markets have come to ask much the same. Johannesburg cannot currently show either, and until someone sets down in writing what is on the floor and what should replace it, the plant stays exposed on three fronts -- a failed inspection, lost market access for Moringa Daily, and batch records that remain manual and therefore error-prone. Capital expenditure approval and procurement both wait on this report: it is the evidence base the engineering and quality teams do not yet have.

Brief

You are the automation consultant Johannesburg has brought in, and your report goes to the plant's engineering and quality management teams together. Survey what is on the floor now and map it onto the ISA-95 pyramid. Weigh PLC against DCS for the upgrade and say which one you would buy. Recommend data transport protocols that hold up through load-shedding, and set out what SAHPRA GMP / GxP compliance will ask of the new system.

Key Requirements

- Describe the ISA-95 automation pyramid (Levels 0-4) and identify where each existing system sits
- Compare PLC and DCS architectures in a structured table, then name the choice you would make here and say why
- Evaluate at least three industrial data transport protocols -- from Modbus, OPC UA, MQTT and WebSockets -- against the resilience load-shedding demands and against the GxP environment they will run in

- Draw a network architecture diagram that runs from field devices up to MES/ERP
- Address SAHPRA GMP requirements (and pan-African AMRH harmonisation context) for computerised systems and data integrity (ALCOA+ principles)
- Write the recommendations so that an engineer and a quality manager can each act on them without a translator

Production Guidelines

1. **Research the ISA-95 framework**: Work through the five-level automation pyramid (Levels 0-4) and write your own definitions, with an example of each taken from capsule / OSD manufacturing rather than from a textbook. L1, L2, L6, L8 and L11 are your starting references: go to L2 (Data Integrity and Cybersecurity) for the ALCOA+ principles and how they apply to computerised systems in a GxP environment, and to L11 (QA / 21 CFR Part 11) for the regulatory framework and the pharmaceutical quality system this plant has to sit inside.
2. **Audit the current state**: Describe the standalone PLC systems at Ubuntu Johannesburg and put each one on an ISA-95 level. The empty levels matter as much as the occupied ones -- there is no group-wide MES at Level 3 and no centralised SCADA at Level 2 -- so name the gaps rather than leaving the reader to infer them.
3. **Build the PLC vs. DCS comparison table**: Use at least 6 comparison criteria; scalability, cost, GxP suitability, vendor lock-in, maintenance complexity and connectivity will get you there. Cite vendor documentation or industry sources wherever you can find them.
4. **Evaluate data transport protocols**: Pick at least three from Modbus, OPC UA, MQTT, and WebSockets, then test each against what this plant has to live with: resilience through load-shedding, local edge processing, GxP audit trail requirements, and limited bandwidth.
5. **Draft the network architecture diagram**: Lay it out in layers, field devices at the bottom through to MES/ERP at the top. Mark the data flows, the protocol you chose at each boundary, and the edge computing layer that carries the site through a load-shedding outage.
6. **Address regulatory requirements**: Read the SAHPRA GMP requirements for computerised systems, the WHO TRS Annex 4 expectations that sit alongside them, the AMRH harmonisation context around both, and the ALCOA+ principles. Then show how your architecture holds data integrity all the way from the field sensor to the batch record.
7. **Write for a dual audience**: Open with an executive summary a quality manager can read on its own, then a technical body with enough detail for an engineer to build from. Define your acronyms, add a glossary, and label the diagrams properly.
8. **Review and refine**: Check the word count (3,000-4,000 words), go back through the six key requirements one at a time, look at the diagrams at the size they will actually be read, and proofread for technical accuracy.

Marking Scheme

Criterion	Weight	Fail (0-39)	Pass (40-59)	Merit (60-69)	Distinction (70-84)	Stretch (85-100)
ISA-95 Pyramid Mapping	20%	Pyramid described inaccurately or generically with no connection to the Johannesburg plant; levels confused or missing.	Pyramid correctly described at a textbook level; some attempt to map the plant's existing systems to levels, but mapping is incomplete or superficial.	All five levels accurately described and existing systems correctly mapped; gaps in the plant's coverage identified for most levels.	Every level mapped and every gap pinned down precisely, with a clear explanation of what capability is missing at each level and why it matters at SAHPRA inspection.	Mapping includes a maturity assessment per level, benchmarks against the public record for comparable African pharma / wellness facilities, and a prioritised gap-closure sequence with dependencies identified.
PLC vs. DCS Comparison	20%	No structured comparison; vague statements about one or both architectures with factual errors.	Comparison table present with at least 4 criteria; correct at a basic level but lacking depth or scenario-specific justification.	Table includes 6+ criteria with accurate technical detail; recommendation clearly stated and justified for the capsule line context.	Recommendation accounts for the plant's specific constraints (legacy PLCs, IT staffing, GxP); includes migration path considerations and total cost of ownership reasoning.	Comparison draws on real vendor specifications and includes a hybrid architecture option; discusses future-proofing for potential expansion into adjacent Ubuntu Naturals dosage forms (tablets, liquids); references peer facility case studies.
Protocol Evaluation	15%	Fewer than three protocols evaluated, or descriptions are factually incorrect; no consideration of the plant's constraints.	Three protocols evaluated with basic descriptions of each; some reference to load-shedding constraints but analysis is generic.	Protocols evaluated with clear suitability scoring against plant-specific criteria (load-shedding resilience, GxP, bandwidth); trade-offs articulated.	Evaluation includes store-and-forward strategies, edge buffering design for load-shedding outages, and specific GxP audit trail implications for each protocol.	Includes a protocol selection decision matrix weighted by plant priorities; proposes a staged protocol migration plan; addresses protocol-level cybersecurity considerations.

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Network Architecture Diagram	15%	No diagram, or diagram is a generic copy from a textbook with no adaptation to the scenario.	Diagram present showing basic layered architecture; some labels but missing data flows or protocol annotations.	Well-structured layered diagram with clear labels, data flow arrows, protocol annotations at each boundary, and the edge computing layer shown.	Diagram is detailed enough to guide implementation; includes redundancy paths, failover for load-shedding outages, and annotations linking to GxP requirements.	Diagram includes a companion logical/physical network view; addresses cybersecurity zones (Purdue model segmentation); shows integration timeline phases overlaid on architecture.
Regulatory and Data Integrity (ALCOA+)	15%	ALCOA+ not mentioned or described incorrectly; no reference to SAHPRA / WHO TRS requirements for computerised systems; no evidence of engagement with L2 (Data Integrity and Cybersecurity) lecture content.	ALCOA+ principles listed and briefly explained; some link to the proposed system but not systematic; basic reference to data integrity concepts from L2.	Each ALCOA+ principle connected to a specific feature of the proposed architecture (e.g., audit trails for Attributable, time-stamping for Contemporaneous); SAHPRA GMP and WHO TRS Annex 4 referenced; demonstrates clear application of L2 data integrity frameworks to the automation upgrade context.	Detailed mapping of ALCOA+ to both the current gaps and proposed system capabilities; specific recommendations for each principle with implementation examples; integrates L2 cybersecurity concepts (access controls, audit trail integrity) with the architecture design.	Includes a compliance risk matrix scoring current vs. proposed state against each ALCOA+ attribute; references recent SAHPRA / WHO / PIC/S inspection findings at comparable facilities; proposes a validation master plan outline; synthesises L2 data integrity and cybersecurity content as cross-cutting requirements across the architecture.
Communication Quality	15%	Poorly structured; no executive summary; technical jargon unexplained; significant grammatical or factual errors.	Readable structure with introduction and conclusion; some jargon defined; writing is functional but unpolished.	Clear executive summary; logical flow; glossary provided; writing is accessible to both engineering and quality audiences.	Dual-audience writing is skilfully handled: quality managers can extract compliance implications while engineers find sufficient technical depth; professional report formatting.	Report reads as a publishable consultancy deliverable; includes a one-page visual summary suitable for board presentation; language is precise, persuasive, and free of errors.

Stretch Goal

An outstanding submission would include a validated-systems lifecycle plan showing how the proposed architecture moves from design qualification through to operational qualification, with a preliminary validation master plan outline. It would benchmark the proposed upgrade against what is on the public record for 2-3 comparable African facilities -- WHO prequalification public inspection reports, published SAHPRA and peer-regulator inspection findings, ISPE or PDA case studies, vendor white papers -- rather than against internal architecture no manufacturer publishes, include a sensitivity analysis of total cost of ownership under different load-shedding-resilience investment scenarios, and propose a phased commissioning strategy that maintains Moringa Daily production continuity during the upgrade.

Submitting This Artefact

What you hand in: Technical report (3,000-4,000 words) with diagrams, comparison tables, and recommendations.

This is a Phase 1 (Foundational) artefact. It builds on programme entry, and the work you do here carries into A7 (the data integrity audit, which applies the ALCOA+ work you do here), A9 (the smart factory process map, which relies on the ISA-95 fluency you build here) -- so it is worth doing properly the first time rather than twice.

This brief is attached to the assignment as `A01_automation_landscape_report.pdf`; you work from it plus the lecture and workshop materials it names. A worked exemplar from another Ubuntu Naturals site is attached too, with the marker's note on it, and `A01_genai_guide.pdf` shows how to build your own submission with a free Claude account.

Name the file `A01_Automation_Landscape_Report_[YourSurname].pdf`, with your own surname in place of `[YourSurname]` -- the portfolio guide uses this convention throughout, and a marker looking for your work will look for that name.

Whatever else it contains, the file needs 11-12pt body text, clear headings, numbered sections, page numbers.